

ARCHAEOLOGY

12. Future

2025 AD – end of time



Let's look into the future and imagine the archaeologists who will one day study us. They will probably recognize our era partly by the layer of plastic and concrete in sediments. Digital data can disappear faster than writing carved in stone, because storage media and formats become obsolete. That is why people create time capsules – containers of objects that are opened after many years. In a school time capsule it is worth putting letters and photos of everyday life, so that people 50 years from now understand our lives. The Voyager Golden Record, sent into space in 1977, contains, among other things, music from various cultures. The Arch Mission project sent a library to the Moon recorded on nickel discs. The Svalbard Global Seed Vault is meant to store seeds for hundreds of years, and the Onkalo repository in Finland to store nuclear waste safely for 100,000 years. How do you warn people 10,000 years from now about such a place? That is a problem of communicating with distant generations. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Voyager Golden Record is a message for future discoverers. Among other things, it contains...**
 - a) soccer scores
 - b) music from various cultures
 - c) advertisements
 - d) the anthem of a single country
2. 👁️ **A time capsule is a container of objects that...**
 - a) is sent by mail
 - b) is thrown away
 - c) is eaten
 - d) is opened after many years
3. 🗉 **Future archaeologists will study our era partly by the layer of...**
 - a) gold
 - b) ice
 - c) plastic and concrete in sediments
 - d) volcanic ash
4. 📡 **Digital data can disappear faster than writing carved in stone because...**
 - a) stone decays quickly
 - b) electricity lasts forever
 - c) storage media and formats become obsolete
 - d) paper lasts forever
5. 🌱 **The Svalbard Global Seed Vault is meant to store seeds...**
 - a) for a year
 - b) until tomorrow
 - c) for hundreds of years
 - d) for a week
6. 🌌 **The Arch Mission project sent a library to the Moon that was...**
 - a) made of paper
 - b) stored only in the cloud
 - c) recorded on nickel discs
 - d) made of wood
7. 💖 **What is worth putting in a school time capsule so that people 50 years from now understand our lives?**
 - a) letters and photos of everyday life
 - b) food
 - c) live animals
 - d) nothing
8. 🌿 **The Onkalo repository in Finland is meant to store nuclear waste safely for...**
 - a) 100,000 years
 - b) 100 years
 - c) 10 years
 - d) 1 year
9. ⭐ **How do you warn people 10,000 years from now about a nuclear waste site? This is a problem of...**
 - a) road traffic
 - b) communicating with distant generations
 - c) the local newspaper
 - d) fashion

PHILOSOPHY

12. Future

2025 AD – end of time



Let's travel in thought into the future, where philosophy asks big questions. Thinking about the future helps people set goals for themselves. Transhumanism is the idea of enhancing humans with technology, and the question "Is it acceptable to create people with selected genes?" belongs to bioethics. Philosophers also ask whether a future artificial intelligence could have consciousness – so far there is no evidence of it. Pope Leo XIV announced that the Church would address the challenges of artificial intelligence. Longtermism is the view that we should care about people in the distant future. Hans Jonas wrote about the "imperative of responsibility": act so that the effects of your actions do not destroy future life on Earth. Whether we have a duty to protect the climate for future generations is a question of intergenerational justice. And according to physics, trillions of years from now the Universe may cool down and sink into darkness. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Transhumanism is the idea of...**
 - a) returning to nature
 - b) enhancing humans with technology
 - c) living without music
 - d) living in trees
2. 👁️ **Philosophers ask whether a future artificial intelligence could have...**
 - a) a color
 - b) a weight
 - c) consciousness
 - d) a taste
3. 🗨️ **Longtermism is the view that we should care about...**
 - a) only today
 - b) only ourselves
 - c) people in the distant future
 - d) only animals
4. ⚖️ **The question “Is it acceptable to create people with selected genes?” belongs to...**
 - a) bioethics
 - b) geography
 - c) astronomy
 - d) cooking
5. ⚡ **Hans Jonas wrote about the “imperative of responsibility”: act so that the effects of your actions...**
 - a) do not destroy future life on Earth
 - b) are quick
 - c) bring profit
 - d) stay hidden
6. 🙋 **Pope Leo XIV announced that the Church would address the challenges of...**
 - a) artificial intelligence
 - b) fashion
 - c) sports
 - d) conquering Mars
7. 💖 **Thinking about the future helps people...**
 - a) forget things
 - b) do nothing
 - c) fear everything
 - d) set goals for themselves
8. 🌿 **Whether we have a duty to protect the climate for future generations is a question of...**
 - a) intergenerational justice
 - b) mathematics
 - c) grammar
 - d) sports
9. ⭐ **According to physics, trillions of years from now the Universe may...**
 - a) cool down and sink into darkness
 - b) turn into sugar
 - c) vanish tomorrow
 - d) become the Earth

HISTORY

12. Future

2025 AD – end of time



Let's look into the future through a historian's eyes. From 2025 to 2100, 75 years will pass – many of today's children will live to see the 22nd century. The year 2066 will mark 1,100 years since the Baptism of Poland, and Poland will celebrate the 200th anniversary of regaining independence in 2118. Future historians will study our times mainly through digital data and internet archives. Some people are already thinking very far ahead: the Clock of the Long Now, being built in Texas, is designed to run for 10,000 years, and John Cage's piece "Organ²/ASLSP" in a church in Halberstadt is scheduled to last until 2640. Climate scientists predict that without action, sea levels could rise by several dozen cm or more by 2100. By the time today's children grow up, many jobs will have changed because of technology. Is history predetermined? Historians answer that the future depends on people's decisions. Remember these numbers – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 John Cage's piece "Organ²/ASLSP" in a church in Halberstadt is scheduled to last until...
 - a) 2640
 - b) 3000
 - c) 2030
 - d) 2100
2. 🕒 The Clock of the Long Now, being built in Texas, is designed to run for...
 - a) 10 years
 - b) 100 years
 - c) 10,000 years
 - d) 1,000 days
3. 📖 Future historians will study our times mainly through...
 - a) papyri
 - b) parchments
 - c) digital data and internet archives
 - d) clay tablets
4. 📅 How many years will pass from 2025 to 2100?

5. 🇵🇱 Poland will celebrate the 200th anniversary of regaining independence in...
 - a) 2028
 - b) 2050
 - c) 2200
 - d) 2118
6. 🇵🇱 The year 2066 will mark 1,100 years since...
 - a) the Constitution of May 3
 - b) the Battle of Grunwald
 - c) the Baptism of Poland
 - d) the Union of Lublin
7. 💖 By the time today's children grow up, many jobs will...
 - a) be banned
 - b) have changed because of technology
 - c) be exactly the same
 - d) have disappeared completely
8. 🌿 Climate scientists predict that without action, sea levels could rise by 2100 by...
 - a) 1 mm
 - b) several dozen cm or more
 - c) several km
 - d) nothing at all
9. ☆ Is history "predetermined"? Historians usually answer that...
 - a) everything is written in advance
 - b) the planets decide
 - c) the future depends on people's decisions
 - d) nothing ever changes

LINGUISTICS

12. Future

2025 AD – end of time



Let's listen to what the future of languages may sound like. Linguists warn that by 2100 the world could lose as many as half of its roughly 7,000 languages. Those working to save endangered languages include UNESCO and local communities. In Poland, the Kashubian language is protected as a regional language. New words appear most often because of new things and phenomena. In the future, AI translators in earbuds are meant to make live conversation across languages possible, and sign languages may be translated by cameras and artificial intelligence. Will learning languages still be needed then? Many experts say yes – language also means culture and relationships. Scientists from Project CETI are trying to understand the language of sperm whales. And how could we communicate with an extraterrestrial civilization? Scientists suggest the language of mathematics and physics. Remember these predictions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏳️ **By 2100, the world could lose as many as...**
 - a) half of its roughly 7,000 languages
 - b) all languages except one
 - c) no languages at all
 - d) only English
2. 👁️ **In the future, AI translators in earbuds are meant to make possible...**
 - a) telepathy
 - b) mind reading
 - c) learning without school
 - d) live conversation across languages
3. 🗨️ **Will learning foreign languages still be needed in the age of AI translators? Many experts say...**
 - a) only for robots
 - b) no – it will disappear
 - c) yes – language also means culture
 - d) only Latin will be needed
4. 📖 **New words appear in a language most often because of...**
 - a) new things and phenomena
 - b) the weather
 - c) bans
 - d) boredom
5. 🗣️ **In the future, sign languages may be translated by...**
 - a) cameras and artificial intelligence
 - b) pigeons
 - c) the telegraph
 - d) radio
6. 🧑 **Those working to save endangered languages include...**
 - a) nobody
 - b) shops
 - c) only the military
 - d) UNESCO and local communities
7. ❤️ **In Poland, the Kashubian language is protected as a...**
 - a) banned dialect
 - b) dead language
 - c) regional language
 - d) foreign language
8. 🍃 **Will animals “talk” to us thanks to AI? Scientists from Project CETI are trying to understand...**
 - a) pet canaries singing
 - b) the language of sperm whales
 - c) only dogs barking
 - d) frogs croaking in Poland
9. 🌟 **How could we communicate with an extraterrestrial civilization? Scientists suggest the language of...**
 - a) Polish
 - b) mathematics and physics
 - c) Latin
 - d) emoji

LITERARY STUDIES

12. Future

2025 AD – end of time



Let's open the books that tried to guess the future. A utopia is a vision of an ideal society, while dystopia – its opposite – describes a bad, frightening future. George Orwell's novel *Nineteen Eighty-Four* warns against total surveillance. In *From the Earth to the Moon* from 1865, Jules Verne described a launch from Florida – more than a century later, *Apollo 11* launched from there. Isaac Asimov formulated the Three Laws of Robotics. In *Summa Technologiae* from 1964, Stanisław Lem predicted, among other things, virtual reality, and in *Solaris* humans encounter an ocean that they are unable to understand. In Frank Herbert's *Dune*, set on a desert planet, the most precious thing is water. Science fiction literature helps people reflect on where technology leads. Remember these books – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In “Summa Technologiae” (1964), Stanisław Lem predicted, among other things...
 - a) horse-drawn trams
 - b) virtual reality
 - c) black-and-white TV
 - d) kerosene lamps
2. 👁 George Orwell's novel “Nineteen Eighty-Four” warns against...
 - a) dragons
 - b) total surveillance
 - c) floods
 - d) aliens
3. 📖 A utopia is a vision of...
 - a) a war
 - b) a monster
 - c) a disaster
 - d) an ideal society
4. ⚖ Dystopia is the opposite of utopia – it describes...
 - a) an ideal world
 - b) a fairy tale
 - c) a vacation
 - d) a bad, frightening future
5. ⚡ Isaac Asimov formulated the Three Laws of...
 - a) Motion
 - b) Traffic
 - c) Robotics
 - d) Chess
6. 🌐 In “From the Earth to the Moon” (1865), Jules Verne described a launch from Florida – more than a century later, the mission launched from there was...
 - a) Sputnik
 - b) the Titanic
 - c) Apollo 11
 - d) the first airplane
7. 💖 In Lem's “Solaris”, humans encounter an ocean that they...
 - a) can drink
 - b) are unable to understand
 - c) can easily study
 - d) can freeze
8. 🌿 Frank Herbert's novel “Dune” is set on a desert planet where the most precious thing is...
 - a) gold
 - b) oil
 - c) water
 - d) salt
9. ☆ Science fiction literature helps people...
 - a) forget about science
 - b) avoid thinking
 - c) reflect on where technology leads
 - d) learn to write neatly

CULTURE AND RELIGION

12. Future

2025 AD – end of time



Let's see what the coming years will bring in culture and religion. The Holy Year 2025 began when the Pope opened the Holy Door of St. Peter's Basilica, and the next ordinary Jubilee Year will fall in 2050. The 2026 Winter Olympics are being held in Italy (Milan–Cortina), and the 2028 Summer Games will be held in Los Angeles. The next World Youth Day is scheduled for 2027 in Seoul. The European Capital of Culture in 2029 is to be a city in Poland. Construction of the Sagrada Família in Barcelona is expected to be completed around 2026–2034. John Cage's piece in Halberstadt is performed so slowly that a single sound lasts up to several years. And in many religions, a person's future after death is described through the hope of eternal life. Remember these dates – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 John Cage's piece in Halberstadt is performed so slowly that a single sound lasts...
 - a) a minute
 - b) up to several years
 - c) an hour
 - d) a second
2. 👁️ Construction of the Sagrada Família in Barcelona is expected to be completed around...
 - a) 2026–2034
 - b) 3000
 - c) 2020
 - d) never
3. 🗨️ The Holy Year 2025 began when the Pope opened...
 - a) a bridge over the Tiber
 - b) the gate of Wawel Castle
 - c) the Holy Door of St. Peter's Basilica
 - d) a new church in Rome
4. 🗓️ After 2025, the next ordinary Jubilee Year in the Church will fall in...
 - a) 2100
 - b) 2050
 - c) 2030
 - d) 2040
5. 🌐 The next World Youth Day is scheduled for 2027 in...
 - a) Kraków
 - b) Paris
 - c) Seoul
 - d) Rio
6. 🏊 The 2028 Olympic Games will be held in...
 - a) Tokyo
 - b) Paris
 - c) Los Angeles
 - d) Warsaw
7. 💖 In many religions, a person's future after death is described through the hope of...
 - a) oblivion
 - b) reincarnation as a stone
 - c) work
 - d) eternal life
8. 🍃 The 2026 Winter Olympics are being held in...
 - a) Norway
 - b) Poland
 - c) Japan
 - d) Italy (Milan–Cortina)
9. ☆ The European Capital of Culture in 2029 is to be a city in...
 - a) France
 - b) Greece
 - c) Poland
 - d) Spain

ECONOMICS AND FINANCE

12. Future

2025 AD – end of time



Let's peek into the wallet of the future. Bulgaria adopted the euro on January 1, 2026, and the EU's central bank is preparing a digital euro – electronic money issued by the central bank. Forecasts say that by 2050 the world will have about 9.5–10 billion people. Because Poland's population is aging, fewer people will be working to fund pensions. Automation may replace many repetitive jobs, so it is worth developing creativity and teamwork. A circular economy is meant to ensure that waste becomes raw material. Economists warn that the costs of climate change may be much higher than the costs of prevention. And saving "for the future" means setting aside part of your money: if you save 10 PLN a week, after 52 weeks you will have 520 PLN. Remember these numbers – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 The EU's central bank is preparing a digital euro, which is...
 - a) new gold coins
 - b) a private cryptocurrency
 - c) electronic money issued by the central bank
 - d) vouchers
2. 👁 Bulgaria adopted the euro on...
 - a) 2040
 - b) 2002
 - c) 2014
 - d) January 1, 2026
3. 🗨 In the future, a circular economy is meant to ensure that...
 - a) there are no factories
 - b) waste becomes raw material
 - c) everything gets thrown away
 - d) there is no trade
4. ⚙ Automation may replace many repetitive jobs in the future, so people should...
 - a) rely only on robots
 - b) develop creativity and teamwork
 - c) stop learning
 - d) stop working
5. 💰 Saving “for the future” means...
 - a) throwing money away
 - b) setting aside part of your money
 - c) always borrowing
 - d) spending everything
6. 👴 Because Poland's population is aging, in the future...
 - a) fewer people will be working to fund pensions
 - b) there will be no seniors
 - c) there will be more children
 - d) pensions will disappear
7. 💜 If you save 10 PLN (Polish zlotys) a week, how much will you have saved after 52 weeks?

8. 🌿 The future costs of climate change may be...
 - a) zero
 - b) lower than today
 - c) much higher than the costs of prevention
 - d) unknown to anyone
9. ☆ Forecasts say that by 2050 the world will have about...
 - a) 5 billion people
 - b) 1 billion people
 - c) 9.5–10 billion people
 - d) 50 billion people

PSYCHOLOGY



12. Future

2025 AD – end of time

Let's talk about how our mind looks toward the future. The question "Who do I want to be when I grow up?" concerns plans and values for the future. Writing a letter to your future self helps you reflect on your goals. The marshmallow test, a delayed gratification study, shows that the ability to wait helps achieve goals in the future. Optimism about the future helps people cope with difficulties. Many young people feel "climate anxiety" – worry about the planet's future – and psychologists say that taking action and community help. Even in a future dominated by technology, relationships with people will still be key to happiness. In future cities, spending time in nature is expected to be possible thanks to parks and green corridors. And music therapy may be supported by AI that chooses music to match the patient's mood. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, music therapy may be supported by AI that...
 - a) records dreams
 - b) plays instead of the doctor
 - c) bans music
 - d) chooses music to match the patient's mood
2. 👁️ “Climate anxiety” is...
 - a) worry about the planet's future
 - b) the title of a movie
 - c) fear of rain
 - d) an animal disease
3. 🗨️ Psychologists say that what helps with anxiety about the future is...
 - a) ignoring the world
 - b) taking action and community
 - c) loneliness
 - d) hiding away
4. ⚖️ The delayed gratification study (the marshmallow test) shows that the ability to wait...
 - a) does not matter
 - b) helps achieve goals in the future
 - c) is always innate
 - d) is harmful
5. ✉️ Writing a letter to your future self is an exercise that helps you...
 - a) reflect on your goals
 - b) avoid planning
 - c) spend money
 - d) forget your dreams
6. 🤖 In a future dominated by technology, relationships with people will be...
 - a) unnecessary
 - b) fully replaced by robots
 - c) still key to happiness
 - d) banned
7. ❤️ Optimism about the future...
 - a) harms your health
 - b) does not exist
 - c) is always naive
 - d) helps people cope with difficulties
8. 🌿 In future cities, spending time in nature is expected to be possible thanks to...
 - a) parks and green corridors
 - b) concrete
 - c) only screens showing forests
 - d) bans on going outside
9. ⭐ The question “Who do I want to be when I grow up?” concerns...
 - a) only money
 - b) nothing
 - c) plans and values for the future
 - d) only work

EDUCATION

12. Future

2025 AD – end of time



Let's peek into the school of the future. According to many experts, it will focus more on teaching thinking and cooperation. The skills of the future include critical thinking and creativity. Project-based learning involves solving problems in groups, and planning their own learning helps students become responsible. An AI learning assistant may adapt the learning pace to each student, while the teacher will increasingly be a guide and mentor. Can a robot replace a teacher? Most educators say no – relationships and values matter. Climate education is meant to teach how to protect the planet. And lifelong learning means that we keep learning as adults too. Remember these predictions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

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1. 🏆 According to many experts, the school of the future will focus more on teaching...
 - a) in silence and solitude
 - b) thinking and cooperation
 - c) only spelling tests
 - d) only dates
2. 👁 In the future, an AI learning assistant may...
 - a) adapt the learning pace to each student
 - b) ban books
 - c) replace parents at home
 - d) abolish recess
3. 🗨 Lifelong learning means that...
 - a) there is no need to learn
 - b) learning ends at school
 - c) only children learn
 - d) we keep learning as adults too
4. ➡ According to many reports, the skills of the future include...
 - a) critical thinking and creativity
 - b) knowledge of Latin
 - c) neat handwriting
 - d) only fast mental arithmetic
5. 🚀 Project-based learning involves...
 - a) listening to lectures
 - b) writing spelling tests
 - c) cramming by heart
 - d) solving problems in groups
6. 🤖 The teacher of the future will increasingly be a...
 - a) guide and mentor
 - b) cashier
 - c) robot
 - d) guard
7. 💖 Planning their own learning helps students...
 - a) become responsible
 - b) forget things
 - c) depend on others
 - d) avoid learning
8. 🌱 In the future, climate education is meant to teach...
 - a) how to protect the planet
 - b) how to destroy forests
 - c) how to ignore science
 - d) how to waste water
9. ☆ Can a robot replace a teacher? Most educators say...
 - a) yes, completely
 - b) no – relationships and values matter
 - c) nobody knows anything
 - d) yes, starting tomorrow

SOCIOLOGY

12. Future

2025 AD – end of time



Let's imagine the society of the future. According to UN forecasts, by 2050 about 2/3 of all people will live in cities. Future cities are meant to be friendly to pedestrians and cyclists, which means they will have more bike paths and fewer cars, and "sponge cities" are meant to retain rainwater. Forecasts say that by 2060 Poland's population will shrink, and aging societies will need more care for seniors. Climate migration is the movement of people because of droughts, floods and rising seas. The UN Sustainable Development Goals were set for the year 2030. Volunteering and helping neighbors will remain important for communities. And concerts may also take place in virtual reality. Remember these forecasts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, concerts may also take place in...
 - a) space only
 - b) water
 - c) virtual reality
 - d) silence
2. 👁 According to UN forecasts, by 2050 the share of people living in cities will be about...
 - a) 2/3 of all people
 - b) everyone
 - c) 1/10 of all people
 - d) no one
3. 🗨 In the future, aging societies will need...
 - a) fewer homes
 - b) fewer doctors
 - c) hospital closures
 - d) more care for seniors
4. ➡ Forecasts say that by 2060 Poland's population will...
 - a) shrink
 - b) double
 - c) stay the same
 - d) grow tenfold
5. ➡ Future cities are meant to be friendly to pedestrians and cyclists, which means they will...
 - a) have more bike paths and fewer cars
 - b) have no sidewalks
 - c) be underground
 - d) have more highways
6. 🌍 Climate migration is the movement of people because of...
 - a) the Olympics
 - b) holidays
 - c) droughts, floods and rising seas
 - d) fashion
7. ❤ In the future, volunteering and helping neighbors will...
 - a) disappear
 - b) remain important for communities
 - c) be replaced by robots
 - d) be banned
8. 🌿 In the future, “sponge cities” are meant to...
 - a) dry out
 - b) float on the sea
 - c) glow
 - d) retain rainwater
9. ☆ The UN Sustainable Development Goals (SDGs) were set for the year...
 - a) 2100
 - b) 2030
 - c) 2050
 - d) 2020

LAW

12. Future

2025 AD – end of time



Let's consider what law awaits us in the future. In Poland, copyright expires 70 years after the author's death, so the works of Bruno Schulz, who died in 1942, are in the public domain. The UN Convention on the Rights of the Child states that children have the right to express their views. The European Union aims to achieve climate neutrality by 2050, and the AI Act bans, among other things, "social scoring" of people. The law must also determine who is responsible for an accident involving a self-driving car. The Outer Space Treaty forbids countries from claiming ownership of the Moon, and Poland signed the Artemis Accords in 2021. Some countries have granted rivers the status of a legal person. The first ecosystem in Europe to be granted rights by law – in Spain in 2022 – was the lagoon called Mar Menor. Remember these laws – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🇵🇱 In Poland, copyright expires 70 years after the author's death. The works of Bruno Schulz (d. 1942) are therefore...
 - a) protected until 2100
 - b) state property
 - c) banned
 - d) in the public domain
2. 🌐 The European Union aims to achieve climate neutrality by...
 - a) 2025
 - b) 2200
 - c) 2100
 - d) 2050
3. 🚀 The Outer Space Treaty forbids countries from...
 - a) launching satellites
 - b) crewed flights
 - c) claiming ownership of the Moon
 - d) exploring space
4. 🇵🇱 Poland signed the Artemis Accords in...
 - a) 2001
 - b) 2011
 - c) 2021
 - d) 2031
5. 🚗 In the future, the law must determine who is responsible for an accident involving...
 - a) a bicycle
 - b) a pedestrian
 - c) a self-driving car
 - d) a horse
6. 🧒 The UN Convention on the Rights of the Child states that children have the right to...
 - a) work in a mine
 - b) express their views
 - c) vote in elections
 - d) drive a car
7. 💖 Among other things, the EU AI Act bans...
 - a) online translators
 - b) "social scoring" of people
 - c) using calculators
 - d) computer games
8. 🌿 Some countries have granted rivers the status of...
 - a) a restricted zone
 - b) a private plot
 - c) a highway
 - d) a legal person
9. 🌟 The first ecosystem in Europe to be granted rights by law (Spain, 2022) was the lagoon called...
 - a) Doñana
 - b) Tagus
 - c) Albufera
 - d) Mar Menor

OTHER (HUMANITIES)

12. Future

2025 AD – end of time



Finally, let's look into a future full of dreams and big events. The 1969 song "In the Year 2525" imagines what humanity will be like hundreds and thousands of years from now. The 2026 FIFA World Cup is being held in the USA, Canada and Mexico, and in 2030 it is to be held in countries including Spain, Portugal and Morocco. The first World Cup was held in 1930, so by 2030, 100 years will have passed. The world's fair Expo 2030 is to be held in Riyadh. The Artemis program plans a permanent base on the Moon, but no human will visit Pluto for a long time, because the journey takes almost a decade. Your dreams about the future are best written down and broken into small steps. And the question of what kind of world we will leave to our children concerns each of us. Remember these plans – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 The song “In the Year 2525” (1969) imagines what humanity will be like...
 - a) a week from now
 - b) hundreds and thousands of years from now
 - c) a year from now
 - d) an hour from now
 2. 👁 Expo 2030, the world's fair, is to be held in...
 - a) Paris
 - b) London
 - c) Riyadh
 - d) Warsaw
 3. 🗨 The 2030 FIFA World Cup is to be held in countries including...
 - a) Australia
 - b) China
 - c) Spain, Portugal and Morocco
 - d) Poland
 4. 🏆 The first FIFA World Cup was held in 1930. How many years will have passed by 2030?
-
5. 🏆 The 2026 FIFA World Cup is being held in...
 - a) Germany
 - b) Brazil
 - c) Qatar
 - d) the USA, Canada and Mexico
 6. 🚀 The Artemis program plans a permanent base in the future...
 - a) on the Moon
 - b) on the Sun
 - c) on Venus
 - d) on Jupiter
 7. 💖 Your dreams about the future are best...
 - a) forgotten
 - b) hidden from everyone
 - c) abandoned
 - d) written down and broken into small steps
 8. 🌿 No human will visit the dwarf planet Pluto for a long time, because the journey takes...
 - a) almost a decade
 - b) a month
 - c) a year
 - d) a day
 9. ☆ The question of what kind of world we will leave to our children concerns...
 - a) each of us
 - b) only scientists
 - c) no one
 - d) only politicians
-

1 b 2 c 3 c 4 1005 d 6 a 7 d 8 a 9 a

A N S W E R S



ARCHITECTURE AND URBAN PLANNING

12. Future

2025 AD – end of time

Let's imagine the architecture of the future. In Saudi Arabia, Jeddah Tower is rising – the world's tallest building under construction, planned to be about 1,000 m tall. Cities of the future are being planned as "smart cities," meaning cities managed using data from sensors. Zero-energy buildings produce as much energy as they use, and "bioclimatic" architecture is meant to make use of sun, wind and greenery. Floating homes and neighborhoods are being built in the Netherlands. The principle of "universal design" says that spaces should serve everyone, including people with disabilities. Your ideal room of the future could adjust its own lighting and temperature. And further out in space: bases on the Moon could be built from 3D-printed regolith, and homes on Mars would have to protect against radiation and cold. Remember these plans – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚀 **Bases on the Moon could be built from...**
 - a) ice from freezers
 - b) bricks from Earth
 - c) wood
 - d) 3D-printed regolith
2. 🏗️ **The world's tallest building under construction, Jeddah Tower, is planned to be about...**
 - a) 1,000 m tall
 - b) 5 km tall
 - c) 500 m tall
 - d) 100 m tall
3. 🏠 **Cities of the future are being planned as “smart cities”, meaning cities...**
 - a) without electricity
 - b) managed using data from sensors
 - c) under water
 - d) without people
4. ⚡ **In the future, zero-energy buildings will...**
 - a) have no roofs
 - b) produce as much energy as they use
 - c) have no windows
 - d) use a lot of gas
5. 🚀 **Homes on Mars would have to protect against...**
 - a) heat waves
 - b) wind from Earth
 - c) radiation and cold
 - d) rain
6. 🌊 **Floating homes and neighborhoods are being built in...**
 - a) the Czech Republic
 - b) Switzerland
 - c) the Netherlands
 - d) Mongolia
7. 💖 **Your ideal room of the future could...**
 - a) be made of paper
 - b) adjust its own lighting and temperature
 - c) have no doors
 - d) have no windows
8. 🌿 **The architecture of the future is meant to be “bioclimatic”, which means it will...**
 - a) make use of sun, wind and greenery
 - b) use a lot of energy
 - c) ignore the climate
 - d) be made of plastic
9. ⭐ **Will future cities be accessible to people with disabilities? The principle of “universal design” says that...**
 - a) stairs are enough
 - b) spaces should serve everyone
 - c) it applies only to new buildings abroad
 - d) it is unnecessary



12. Future

2025 AD – end of time

Let's look into the electronics of the future. Brain–computer interfaces may allow paralyzed people to speak and write using their thoughts, but implants and electronics inside the body raise questions about the privacy of health data. Solid-state batteries are expected to charge faster. A quantum computer may be able to quickly solve certain very hard problems, which is why post-quantum cryptography is needed – quantum computers could break today's encryption. Smart grids will make it possible to share electricity between homes. In factories, exoskeletons will help workers lift heavy loads without injury. Biodegradable electronics are meant to break down after use. And should AI make decisions about people without oversight? Most experts believe a human should have the final say. Remember these inventions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎧 In the future, brain–computer interfaces may allow paralyzed people to...
 - a) fly
 - b) go without sleep
 - c) speak and write using their thoughts
 - d) read other people's minds
2. 👁 In the future, solid-state batteries are expected to...
 - a) charge faster
 - b) work without electricity
 - c) glow at night
 - d) weigh a ton
3. 🗨 In the future, a quantum computer may be able to...
 - a) cook
 - b) read minds
 - c) replace all computers
 - d) quickly solve certain very hard problems
4. ⚡ Post-quantum cryptography is needed because quantum computers could...
 - a) remove bacteria
 - b) break today's encryption
 - c) switch off the Sun
 - d) fix the internet
5. 🦾 In the factories of the future, exoskeletons will help workers...
 - a) sleep
 - b) fly
 - c) lift heavy loads without injury
 - d) read
6. 🌐 Smart grids will make it possible to...
 - a) share electricity between homes
 - b) switch off power forever
 - c) produce gasoline
 - d) freeze water
7. ❤ Implants and electronics inside the body raise questions about...
 - a) shoe weight
 - b) height
 - c) skin color
 - d) the privacy of health data
8. 🌱 In the future, biodegradable electronics are meant to...
 - a) last forever
 - b) glow forever
 - c) poison the soil
 - d) break down after use
9. ☆ Should AI make decisions about people without human oversight? Most experts believe that...
 - a) no one should decide
 - b) a human should have the final say
 - c) robots should decide
 - d) yes, always

COMPUTER SCIENCE AND TELECOMMUNICATIONS

12. Future

2025 AD – end of time



Let's travel to the internet of the future. The 6G network is expected to launch after 2030 and be even faster than 5G. Satellite internet is meant to reach every place on Earth, and a quantum internet is meant to allow unbreakable encryption of connections. An interplanetary internet will have delays, because a signal traveling from Mars to Earth needs between 3 and 22 minutes. So-called "artificial general intelligence" (AGI) is a hypothetical AI that could do everything a human can. AI data centers consume huge amounts of electricity and cooling water. A "digital footprint" is the information you leave online, and to stay safe online it is worth using strong passwords and watching out for scams. Can AI write books and paint? The more important question is how to use it wisely and honestly. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The 6G network is expected to launch after 2030 and be...**
 - a) radio-only
 - b) slower than 3G
 - c) even faster than 5G
 - d) military-only
2. 🌐 **In the future, satellite internet is meant to reach...**
 - a) nowhere
 - b) only Europe
 - c) only cities
 - d) every place on Earth
3. 🗨️ **So-called “artificial general intelligence” (AGI) is a hypothetical AI that...**
 - a) only plays chess
 - b) only does calculations
 - c) could do everything a human can
 - d) can never exist
4. 🚀 **An interplanetary internet will have delays, because a signal traveling from Mars to Earth needs...**
 - a) exactly one hour
 - b) one second
 - c) between 3 and 22 minutes
 - d) a week
5. ⚡ **To stay safe on the internet of the future, it is worth...**
 - a) never updating your devices
 - b) using strong passwords and watching out for scams
 - c) giving your passwords to everyone
 - d) clicking every link
6. 🧑‍🔬 **In the future, a quantum internet is meant to allow...**
 - a) free electricity
 - b) unbreakable encryption of connections
 - c) telepathy
 - d) faster games
7. 💖 **A “digital footprint” is...**
 - a) a fingerprint
 - b) the information you leave online
 - c) a photo in an album
 - d) a shoe print
8. 🌿 **AI data centers consume huge amounts of...**
 - a) electricity and cooling water
 - b) gasoline
 - c) wood
 - d) sand
9. ⭐ **Can AI write books and paint pictures? The question for the future is rather:**
 - a) how to replace people
 - b) how to ban it everywhere
 - c) how to use it wisely and honestly
 - d) how to forget about art

BIOMEDICAL ENGINEERING

12. Future

2025 AD – end of time



Let's look into the lab of future medicine. Organs 3D-printed from a patient's own cells could avoid being rejected by the body, and transplants from genetically modified pigs may shorten transplant waiting lists. Nanorobots are meant to deliver drugs to diseased cells. Gene therapies may cure diseases that are hereditary. Bionic prostheses are meant to give patients a sense of touch, and cochlear implants to reproduce music even more faithfully. Telemedicine will allow doctors to examine patients remotely. Astronauts flying to Mars will need protection from cosmic radiation. And should enhancing humans with technology be available to everyone? That is a question of fairness and equal access. Remember these visions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **Future cochlear implants are meant to...**
 - a) replace eyes
 - b) cure a runny nose
 - c) play music by themselves
 - d) reproduce music even more faithfully
2. 👁️ **Organs 3D-printed from a patient's own cells could...**
 - a) change color
 - b) fail to work
 - c) be made of plastic
 - d) avoid being rejected by the body
3. 🗨️ **In the medicine of the future, nanorobots are meant to...**
 - a) feed the patient
 - b) clean hospitals
 - c) deliver drugs to diseased cells
 - d) operate without a doctor
4. 🐷 **In the future, organ transplants from genetically modified pigs may...**
 - a) shorten transplant waiting lists
 - b) eliminate hospitals
 - c) change nothing
 - d) replace all medicines
5. 🦾 **Bionic prostheses of the future are meant to give patients...**
 - a) telepathy
 - b) superhuman strength always
 - c) a sense of touch
 - d) wings
6. 🩺 **In the future, telemedicine will allow doctors to...**
 - a) treat without knowledge
 - b) send medicine by pigeon
 - c) examine patients remotely
 - d) not work
7. 💖 **In the future, gene therapies may cure diseases that are...**
 - a) all without exception
 - b) only in animals
 - c) hereditary
 - d) only colds
8. 🌿 **Astronauts flying to Mars will need protection from...**
 - a) rain
 - b) street noise
 - c) smog
 - d) cosmic radiation
9. ⭐ **Should enhancing humans with technology be available to everyone? This is a question of...**
 - a) fairness and equal access
 - b) the weather
 - c) sports
 - d) fashion

CHEMICAL ENGINEERING

12. Future

2025 AD – end of time



Let's look into the chemical engineering of the future. Direct air capture of CO₂ (DAC) is meant to reduce the amount of greenhouse gases, and artificial photosynthesis to produce fuel from light, water and CO₂. Synthetic fuels are produced from CO₂, water and renewable energy. Green hydrogen may replace coal in steelworks. Bioplastics may replace oil-based plastic because they come from plants and break down, and chemical recycling involves breaking plastic down into molecules. A global treaty on plastic pollution is being negotiated by UN member states. One challenge is removing PFAS "forever chemicals," because they build up in nature and in living things. And an everyday decision of yours that will help the chemistry of the future is sorting your waste. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In the future, direct air capture of CO₂ (DAC) is meant to...
 - a) freeze the seas
 - b) increase smog
 - c) produce perfume
 - d) reduce the amount of greenhouse gases
2. 👁 In the future, green hydrogen may replace coal in...
 - a) schools
 - b) steelworks
 - c) gardens
 - d) bakeries only
3. 📄 Synthetic fuels (e-fuels) are produced from...
 - a) wood
 - b) coal
 - c) CO₂, water and renewable energy
 - d) crude oil
4. 🔄 In the future, plant-based bioplastics may replace oil-based plastic because they...
 - a) are always cheaper
 - b) never decompose
 - c) are heavier
 - d) come from plants and break down
5. ✨ Chemical recycling of plastic involves...
 - a) burying it
 - b) dumping it in the sea
 - c) breaking plastic down into molecules
 - d) burning it
6. 🌐 A global treaty on plastic pollution is being negotiated by...
 - a) no one
 - b) only companies
 - c) UN member states
 - d) only Poland
7. ❤ An everyday decision of yours that will help the chemistry of the future is...
 - a) buying disposable items
 - b) burning plastic
 - c) sorting your waste
 - d) throwing batteries in the trash
8. 🌱 In the future, artificial photosynthesis is meant to...
 - a) produce fuel from light, water and CO₂
 - b) color leaves
 - c) stop the Sun
 - d) replace plants entirely
9. ⭐ One challenge for the chemistry of the future is removing PFAS “forever chemicals”, because they...
 - a) taste sweet
 - b) disappear on their own
 - c) build up in nature and in living things
 - d) are too expensive to make

CIVIL ENGINEERING AND TRANSPORT

**12. Future**

2025 AD – end of time

Let's set off on the journeys of the future. The Artemis II mission is to take people around the Moon, and the Lunar Gateway station is meant to orbit the Moon. A trip to Mars takes about 7–9 months because the planets are very far apart. Should humans settle Mars? Supporters talk about the survival of our species, while opponents say we must take care of Earth first. A trip to space as a tourist is possible even today, but at a very high price. On Earth, high-speed rail is meant to connect Warsaw with Łódź and other cities via the CPK hub, and Rail Baltica will link Poland with Lithuania, Latvia and Estonia. Electric airplanes will first serve short routes. Wind-assisted sailing ships are also making a comeback, because they save fuel and cut emissions. Remember these journeys – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **The Artemis II mission is to take people...**
 - a) to Jupiter
 - b) to Mars
 - c) around the Moon
 - d) to the Sun
2. 👁️ **In the future, high-speed rail in Poland is meant to connect...**
 - a) Warsaw with Łódź and other cities via the CPK hub
 - b) Gdańsk with Mars
 - c) Zakopane with Paris by tunnel
 - d) Kraków with Tokyo
3. 🗞️ **In the future, electric airplanes will first serve...**
 - a) short routes
 - b) space flights
 - c) intercontinental flights
 - d) heavy cargo transport
4. 🚀 **A trip to Mars takes about 7–9 months because...**
 - a) rockets are deliberately slow
 - b) Mars is hiding
 - c) you have to wait for good weather
 - d) the planets are very far apart
5. 🚆 **Rail Baltica is a railway under construction linking Poland with...**
 - a) the Czech Republic
 - b) Ukraine
 - c) Germany
 - d) Lithuania, Latvia and Estonia
6. 🌕 **The Lunar Gateway station is meant to orbit...**
 - a) the Moon
 - b) Mars
 - c) the Sun
 - d) Earth in low orbit
7. 💖 **Your first trip to space as a tourist would be possible today...**
 - a) at a very high price
 - b) for free
 - c) by bus
 - d) by train
8. 🍃 **Wind-assisted sailing ships are making a comeback because...**
 - a) they are faster than airplanes
 - b) they save fuel and cut emissions
 - c) they need no crew
 - d) they are fashionable
9. ⭐ **Should humans settle Mars? Supporters talk about the survival of our species, while opponents say that...**
 - a) it is too warm there
 - b) it is too cheap
 - c) we must take care of Earth first
 - d) Mars is too close

1. c 2. a 3. a 4. d 5. d 6. a 7. a 8. b 9. c

ANSWERS

MATERIALS ENGINEERING

12. Future

2025 AD – end of time



Let's get to know the materials of the future. Glass that darkens by itself in sunlight is electrochromic glass. Acoustic metamaterials may silence noise in cities. The clothes of the future may measure your pulse and keep you warm, and thanks to self-healing materials your phone could "heal" scratches on its own screen. Concrete that absorbs CO₂ is meant to help with protecting the climate. Room-temperature superconductors, if they were ever created, would make it possible to transmit electricity without losses. A space elevator would need a cable made of a material hundreds of times stronger than steel. Recycling rare earth metals from electronics will be important because the resources are limited. And when designing products, we need to think about what happens to them after they wear out. Remember these materials – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, acoustic metamaterials may...
 - a) change your voice
 - b) listen to thoughts
 - c) silence noise in cities
 - d) play music by themselves
2. 👁️ Glass that darkens by itself in sunlight is...
 - a) mirror glass
 - b) stained glass
 - c) electrochromic glass
 - d) ordinary window glass
3. 🗨️ Room-temperature superconductors, if they were ever created, would make it possible to...
 - a) cook food
 - b) dye fabrics
 - c) transmit electricity without losses
 - d) freeze water
4. 🚀 A space elevator would need a cable made of a material that is...
 - a) made of paper
 - b) made of rubber
 - c) made of cotton
 - d) hundreds of times stronger than steel
5. 🧑‍🚀 The clothes of the future may...
 - a) fly
 - b) talk to you
 - c) measure your pulse and keep you warm
 - d) wash themselves every night
6. ♻️ Recycling rare earth metals from electronics will be important because...
 - a) nobody uses them
 - b) the metals are useless
 - c) there are endless amounts of them
 - d) the resources are limited
7. 💖 In the future, self-healing materials may mean that your phone will...
 - a) change shape
 - b) last forever
 - c) call people by itself
 - d) "heal" scratches on its own screen
8. 🌿 Concrete that absorbs CO₂ is meant to help with...
 - a) heating
 - b) protecting the climate
 - c) irrigation
 - d) building sandcastles
9. ⭐ When designing the products of the future, we need to think about what happens to them...
 - a) in the store
 - b) in advertising
 - c) after they wear out
 - d) in the box

MECHANICAL ENGINEERING



12. Future

2025 AD – end of time

Let's meet the machines of the future. Humanoid robots are meant to help in homes and with elderly care, and in agriculture robots may weed fields and pick fruit. Self-driving cars are classified into levels from 0 to 5 – level 5 means fully driverless operation. Delivery drones may deliver packages faster on short routes. Underwater robots will help explore and protect the oceans. On the Moon, robots are meant to build bases before humans arrive, and the European Space Agency's Rosalind Franklin rover is meant to search Mars for traces of ancient life. If you want to build robots, it is worth studying math, physics and programming. Can a robot be a friend? Psychologists point out that relationships with people are irreplaceable. Remember these machines – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏠 In the future, humanoid robots are meant to help...
 - a) in homes and with elderly care
 - b) only with singing anthems
 - c) with prayer
 - d) with voting
2. 👁 The European Space Agency's Rosalind Franklin rover is meant to search Mars for...
 - a) drinking water for humans
 - b) gold
 - c) living aliens
 - d) traces of ancient life
3. 🚗 Self-driving cars are classified into levels from 0 to 5, where level 5 means...
 - a) no automation
 - b) parking only
 - c) fully driverless operation
 - d) cruise control only
4. 🚚 In the future, delivery drones may deliver packages...
 - a) without electricity
 - b) to Mars
 - c) faster on short routes
 - d) only at night
5. 🌾 In the agriculture of the future, robots may...
 - a) cook soup
 - b) weed fields and pick fruit
 - c) write poems
 - d) dig for gold
6. 🌕 In the future, robots on the Moon are meant to...
 - a) build bases before humans arrive
 - b) sell souvenirs
 - c) raise cows
 - d) play soccer
7. ❤ If you want to build robots in the future, it is worth studying...
 - a) only history
 - b) math, physics and programming
 - c) nothing
 - d) only singing
8. 🌿 In the future, underwater robots will help...
 - a) pollute the sea
 - b) build beaches
 - c) catch all the fish
 - d) explore and protect the oceans
9. ⭐ Can a robot be someone's friend? Psychologists point out that...
 - a) friendship is unnecessary
 - b) people are unnecessary
 - c) relationships with people are irreplaceable
 - d) robots are better than people



12. Future

2025 AD – end of time

Let's talk about the energy of the future. Scientists say that to limit warming, by around 2050 we must reach net-zero emissions. Poland's first nuclear power plant is expected to start operating in the 2030s, and small modular reactors (SMRs) are small reactors built in series. Fusion power plants, if they are ever built, will provide lots of energy without emissions. By 2040, offshore wind farms in the Baltic Sea are expected to provide Poland with a dozen or so GW of capacity, and shared European transmission grids will make it possible to send green energy between countries. Pumped-storage power plants, such as Żarnowiec, store energy because they pump water uphill and release it later. Asteroid mining is the idea of extracting metals in space. And you can save energy by switching devices off standby. Remember these plans – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Fusion power plants, if they are ever built, will...**
 - a) require coal
 - b) be noisy and dirty
 - c) provide lots of energy without emissions
 - d) work only during the day
2. 👁 **Small modular reactors (SMRs) are...**
 - a) batteries
 - b) small reactors built in series
 - c) reactors inside phones
 - d) wind turbines
3. 🗨 **Poland's first nuclear power plant is expected to start operating...**
 - a) in 1990
 - b) in the 2030s
 - c) in 2025
 - d) in 2100
4. ⚡ **By 2040, offshore wind farms in the Baltic Sea are expected to provide Poland with...**
 - a) nothing
 - b) a dozen or so GW of capacity
 - c) 1 W
 - d) all the energy in the world
5. ✨ **Asteroid mining is the idea of...**
 - a) extracting metals in space
 - b) building mines on Earth
 - c) catching comets
 - d) burying asteroids
6. 🌍 **In the future, shared European transmission grids will make it possible to...**
 - a) send green energy between countries
 - b) produce coal
 - c) close borders
 - d) switch off the power
7. ❤ **One way you can save energy for the future is...**
 - a) keeping the fridge open
 - b) leaving the lights on
 - c) switching devices off standby
 - d) heating with the window open
8. 🌿 **Pumped-storage power plants, such as Żarnowiec, store energy because they...**
 - a) freeze water
 - b) burn water
 - c) pump water uphill and release it later
 - d) do nothing
9. ☆ **Scientists say that to limit warming, by around 2050 we must...**
 - a) increase emissions
 - b) burn more coal
 - c) reach net-zero emissions
 - d) do nothing

ASTRONOMY



12. Future

2025 AD – end of time

Let's look up at the sky of the future. Around 2029–2030 the Extremely Large Telescope (ELT) in Chile is expected to begin observing, and for 10 years the Vera C. Rubin Observatory is meant to image the whole sky every few nights. In 2029 the asteroid Apophis will pass Earth at a distance of about 32,000 km. Halley's Comet will return close to the Sun in 2061. The next total solar eclipse visible from Poland will not occur until 2135, and because the Moon is moving away from Earth, in the distant future total solar eclipses will end. In about a billion years the Sun will become brighter and the oceans on Earth will evaporate. In about 4.5 billion years the Milky Way will collide with the Andromeda Galaxy, and in about 5 billion years the Sun will become a red giant. Will we find life beyond Earth? Future missions are looking for it on places such as the moons of Jupiter and Saturn. Remember these dates – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The Extremely Large Telescope (ELT) in Chile is expected to begin observing around...**
 - a) 2100
 - b) 2020
 - c) 2029–2030
 - d) 1999
2. 👁️ **Halley's Comet will return close to the Sun in...**
 - a) 2100
 - b) 2061
 - c) 2026
 - d) 2030
3. 🗨️ **In about 4.5 billion years, the Milky Way will collide with...**
 - a) the Andromeda Galaxy
 - b) the Sun
 - c) the Moon
 - d) Mars
4. 📏 **The Moon is moving away from Earth by about 3.8 cm per year, so in the distant future...**
 - a) the Moon will look bigger
 - b) total solar eclipses will end
 - c) the Moon will vanish tomorrow
 - d) the Moon will fall down
5. ⚡ **In 2029, the asteroid Apophis will pass Earth at a distance of...**
 - a) 1 km
 - b) 100 million km
 - c) about 32,000 km
 - d) as far as the Moon
6. 🧑🏫 **For 10 years, the Vera C. Rubin Observatory in Chile is meant to...**
 - a) look only at the Moon
 - b) image the whole sky every few nights
 - c) search for gold
 - d) do nothing at all
7. 💖 **In about a billion years the Sun will become brighter, and on Earth...**
 - a) nothing will change
 - b) there will be more water
 - c) the oceans will evaporate
 - d) it will get colder
8. 🌿 **The next total solar eclipse visible from Poland will occur in...**
 - a) 2026
 - b) 2050
 - c) 2030
 - d) 2135
9. ☆ **In about 5 billion years, when it runs out of hydrogen in its core, the Sun will...**
 - a) turn into a black hole
 - b) go dark in a single day
 - c) explode as a supernova
 - d) become a red giant

MATHEMATICS



12. Future

2025 AD – end of time

Let's look into the mathematics of the future. Exponential growth means that a quantity grows faster and faster: if a bacterial population doubles every hour, after 5 hours it will have grown 32 times. Climate models are equations describing air and oceans. Can the future be predicted? In chaotic systems, such as the weather, only for a short time. Statistics will help medical teams predict the risk of disease, and mathematics will help compress music so that it takes up less space. AI assistants are meant to help mathematicians check and discover proofs. Among the seven Millennium Prize Problems, those still unsolved include the Riemann hypothesis. And the ability to think mathematically will be useful in almost every profession. Remember these numbers – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, mathematics will help compress music so that it...
 - a) sounds worse
 - b) is quieter
 - c) takes up less space
 - d) is louder
2. 👁 Among the seven Millennium Prize Problems (a million dollars each), those still unsolved include...
 - a) the Riemann hypothesis
 - b) the Pythagorean theorem
 - c) the multiplication table
 - d) adding fractions
3. 🗨 In the future, AI assistants are meant to help mathematicians...
 - a) check and discover proofs
 - b) abolish mathematics
 - c) count only to 10
 - d) replace schools
4. ➡ If a bacterial population doubles every hour, by how many times will it have grown after 5 hours?

5. 🌤 Future climate models are...
 - a) fortune-telling
 - b) drawings of the sky
 - c) equations describing air and oceans
 - d) photos of clouds
6. 🏥 In the future, statistics will help medical teams...
 - a) predict the risk of disease
 - b) replace patients
 - c) count pills by hand
 - d) draw diagnoses at random
7. ❤ In the future, the ability to think mathematically will be useful...
 - a) only to robots
 - b) never
 - c) in almost every profession
 - d) only to mathematicians
8. 🌱 Exponential growth, for example in emissions or population, means that the quantity...
 - a) grows by equal amounts
 - b) shrinks
 - c) stays the same
 - d) grows faster and faster
9. ☆ Can the future be predicted mathematically? In chaotic systems, such as the weather, it can be predicted...
 - a) not at all
 - b) only for a short time
 - c) exactly, forever
 - d) for thousands of years

OTHER (TECHNOLOGY)



12. Future

2025 AD – end of time

Finally, let's imagine the everyday technology of the future. Holograms at concerts already allow "performances" by artists who have died. 3D-printed food may help astronauts and sick people. The virtual assistant of the future is meant to understand context and emotions in conversation, and AI translators in earbuds will make travel and friendships between nations easier. Augmented reality glasses may display information over the real world. Electronics are being designed to be easy to repair and recycle. Satellites that detect illegal logging help nature. The question of whether everything technically possible is also good is a question of ethics. And regardless of technology, curiosity and learning are worth developing. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎭 **Holograms at concerts already allow “performances” by...**
 - a) artists who have died
 - b) no one
 - c) only robots
 - d) only animals
2. 👁 **In the future, 3D-printed food may help...**
 - a) astronauts and sick people
 - b) farmers sow seeds
 - c) make cooks disappear
 - d) fish
3. 🗨 **The virtual assistant of the future is meant to understand...**
 - a) context and emotions in conversation
 - b) only numbers
 - c) nothing
 - d) only commands
4. ⚖ **So that the electronics of the future do not litter the planet, they are being designed to be...**
 - a) single-use
 - b) made of lead
 - c) impossible to take apart
 - d) easy to repair and recycle
5. 🕶 **In the future, augmented reality (AR) glasses may...**
 - a) change the weather
 - b) replace eyes
 - c) switch off the Sun
 - d) display information over the real world
6. 🗣 **In the future, AI translators in earbuds will make it easier to have...**
 - a) travel and friendships between nations
 - b) forgotten languages
 - c) arguments
 - d) loneliness
7. 💖 **What skill is worth developing for the future, regardless of technology?**
 - a) only memory
 - b) only gaming
 - c) only fast typing
 - d) curiosity and learning
8. 🌿 **Future technologies can help nature – for example, satellites that detect...**
 - a) illegal logging
 - b) the taste of fruit
 - c) animals' dreams
 - d) songbirds
9. ⭐ **The question “Is everything that is technically possible also good?” is a question of...**
 - a) cooking
 - b) sports
 - c) mathematics
 - d) ethics

BIOLOGY

12. Future

2025 AD – end of time



Let's look into the future of the living world. Rachel Carson warned that if we do not protect birds, spring may be quieter – without birdsong. Invasive species may spread faster because the climate is warming. Gene and seed banks are meant to protect the diversity of species. Wildlife crossings over highways help animals move around safely, and rewilding involves, among other things, returning large animals to the wild, such as bison and beavers. Coral reefs may survive thanks to breeding resistant corals. De-extinction – bringing extinct species back – is controversial, because many believe it is better to protect living species. You can help pollinating insects too: just plant native flowers. Do humans have the right to decide the future of other species? Many thinkers believe we have a duty to protect them. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 If we do not protect birds, spring in the future may be...
 - a) quieter – without birdsong
 - b) louder
 - c) warmer
 - d) brighter
2. 👁 In the future, gene and seed banks are meant to protect...
 - a) only people
 - b) only money
 - c) the diversity of species
 - d) only stones
3. 🗨 “De-extinction” – bringing extinct species back – is controversial because...
 - a) it is better to protect living species
 - b) nobody wants it
 - c) it is easy
 - d) it is too cheap
4. ⚖ In the future, invasive species may spread faster because...
 - a) it is getting colder
 - b) the climate is warming
 - c) birds do not fly
 - d) there is no trade
5. 🦋 Wildlife crossings over highways are meant to help animals...
 - a) move around safely
 - b) avoid forests
 - c) drive faster
 - d) find people
6. 🦋 Rewilding – restoring wilderness – involves, among other things...
 - a) paving rivers with concrete
 - b) building cities
 - c) returning large animals to the wild
 - d) keeping animals in cages
7. 💖 What can you do for pollinating insects in the future?
 - a) cut down trees
 - b) plant native flowers
 - c) mow the grass every day
 - d) use more pesticide sprays
8. 🌿 In the future, coral reefs may survive thanks to...
 - a) bleaching
 - b) breeding resistant corals
 - c) burying them
 - d) artificial flowers
9. ⭐ Do humans have the right to decide the future of other species? Many thinkers believe that...
 - a) we have a duty to protect them
 - b) species do not matter
 - c) we have no duties at all
 - d) it is only a matter for scientists

CHEMISTRY

12. Future

2025 AD – end of time



Let's look into the chemistry of the future. Forecasts say that if emissions do not fall, by 2100 the temperature could rise by more than 2–3 °C, and the level of CO₂ in the atmosphere will depend mainly on human decisions about energy. The oceans may become more acidic. There is good news too: the ozone layer is forecast to recover by around 2066, thanks to the ban on CFCs. When burned, hydrogen as a fuel of the future produces only water. Chemists are working on batteries made with sodium, which is cheap and abundant. In dry regions, drinking water will be obtained from sources such as desalinated seawater and the air. Composting kitchen waste gives fertile soil instead of garbage. And if you become a chemist of the future, you can help purify water and create medicines. Remember these forecasts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, the level of CO₂ in the atmosphere will depend mainly on...
 - a) volcanoes on Mars
 - b) human decisions about energy
 - c) music
 - d) the Moon
2. 👁 In the future, the oceans may become more...
 - a) sweet
 - b) permanently cold
 - c) acidic
 - d) thick like honey
3. 🗨 When burned, hydrogen as a fuel of the future produces only...
 - a) smoke
 - b) CO₂
 - c) ash
 - d) water
4. ➡ Can the ozone hole be restored to its pre-1980s state? Forecasts say the ozone layer...
 - a) will never recover
 - b) is already perfect
 - c) will disappear completely
 - d) will recover by around 2066
5. 🌱 In the future, composting kitchen waste is a way to get...
 - a) fertile soil instead of garbage
 - b) polluted water
 - c) more plastic
 - d) smog
6. 👨🔬 Chemists from many countries are working on batteries made with...
 - a) diamonds
 - b) gold
 - c) silver
 - d) sodium, which is cheap and abundant
7. ❤ Will you become a chemist of the future? Chemistry can help, for example, to...
 - a) do nothing
 - b) purify water and create medicines
 - c) only blow things up
 - d) only cook
8. 🍃 In the future, drinking water in dry regions will be obtained from sources such as...
 - a) fruit juices
 - b) desalinated seawater and the air
 - c) ice from the Moon
 - d) meteor showers
9. ⭐ Forecasts say that if emissions do not fall, by 2100 the temperature could rise by...
 - a) nothing at all
 - b) more than 2–3 °C
 - c) 0.1 °C
 - d) 20 °C

PHYSICS

12. Future

2025 AD – end of time



Let's travel into the very distant future that physics predicts. In about a billion years, the Sun's increasing brightness will raise the temperature on Earth. In about 5 billion years the Sun will become a red giant and may swallow Mercury and Venus. At the end of its life the Sun will become a white dwarf, and white dwarfs cool down over time and hypothetically become black dwarfs. The Universe is expanding faster and faster, so distant galaxies will disappear from our view. The expansion could end in a "Big Rip" if dark energy keeps getting stronger – that is just one hypothesis. According to Hawking, after an unimaginably long time black holes will evaporate. The "heat death" of the Universe means a state in which energy is spread out evenly. And much sooner, the Einstein Telescope detector is planned to study gravitational waves. Remember these predictions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **In about 5 billion years, the Sun will become a red giant. At that point it...**
 - a) will go out immediately
 - b) may swallow Mercury and Venus
 - c) will turn into a comet
 - d) will turn blue
2. 👁 **At the end of its life, the Sun will become...**
 - a) a supernova
 - b) a comet
 - c) a white dwarf
 - d) a black hole
3. 🗨 **Over time, white dwarfs cool down and (hypothetically) become...**
 - a) black dwarfs
 - b) suns
 - c) red giants
 - d) planets
4. 📏 **The Universe is expanding faster and faster, so in the future distant galaxies will...**
 - a) stop moving
 - b) disappear from our view
 - c) become brighter
 - d) come closer
5. ⚡ **The expansion of the Universe could end in a “Big Rip” if...**
 - a) gravity wins
 - b) the Sun goes out
 - c) nothing changes
 - d) dark energy keeps getting stronger
6. 🎧 **In the future, the international Einstein Telescope detector will study...**
 - a) musical sounds
 - b) the weather
 - c) gravitational waves
 - d) earthquakes
7. 💖 **According to Hawking, after an unimaginably long time black holes will...**
 - a) turn into stars
 - b) disappear tomorrow
 - c) evaporate
 - d) keep growing forever
8. 🍃 **In about a billion years, the Sun's increasing brightness will...**
 - a) change nothing
 - b) raise the temperature on Earth
 - c) cool Earth down
 - d) stop Earth's rotation
9. ☆ **The “heat death” of the Universe means a state in which...**
 - a) everything catches fire
 - b) a new Earth forms
 - c) energy is spread out evenly
 - d) the Big Bang returns

ENVIRONMENT

12. Future

2025 AD – end of time



Let's look at what the environment of the future may be like. Scientists forecast that in a few decades Poland may have less snow in winter, and by 2100 the glaciers of the Alps may mostly disappear. The level of the Baltic Sea will rise, threatening the coast. Climate tipping points are thresholds beyond which changes become irreversible. The "30×30" target states that by 2030, protection will cover 30% of land and sea. To protect future forests from drought, foresters are planting a mix of species instead of just one. Planting a tree today brings the greatest benefits a few decades from now. You can reduce your ecological footprint, for example, by riding a bike. Will Earth still be a good place to live in 100 years? Scientists say it depends on the decisions we make today. Remember these forecasts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 In a few decades, Poland may have...
 - a) less snow in winter
 - b) more glaciers
 - c) less sunshine
 - d) permanently colder weather
2. 🗞 By 2100, the glaciers of the Alps may...
 - a) suddenly turn into freshwater lakes
 - b) mostly disappear
 - c) stay the same
 - d) grow
3. 📌 Climate tipping points are thresholds beyond which changes...
 - a) stop
 - b) do not matter
 - c) become irreversible
 - d) reverse themselves
4. 🌳 Planting a tree today brings the greatest benefits...
 - a) tomorrow
 - b) a few decades from now
 - c) never
 - d) in a week
5. 🌳 To protect future forests from drought, foresters are planting...
 - a) a mix of species instead of just one
 - b) nothing
 - c) only palm trees
 - d) only spruces
6. 🌳 The “30×30” target states that by 2030, protection will cover...
 - a) 100% of land
 - b) nothing
 - c) 3% of land
 - d) 30% of land and sea
7. ❤ In the future, you can reduce your “ecological footprint”, for example, by...
 - a) throwing away more
 - b) riding a bike
 - c) buying more
 - d) flying more
8. 🌊 In the future, the level of the Baltic Sea...
 - a) will not change
 - b) will rise, threatening the coast
 - c) will drop to zero
 - d) will freeze forever
9. ☆ Will Earth still be a good place to live in 100 years? Scientists say that...
 - a) it certainly will not
 - b) it certainly will, with no effort
 - c) it depends on the decisions we make today
 - d) it does not matter

1. a 2. b 3. c 4. b 5. a 6. d 7. b 8. b 9. c

ANSWERS

HEALTH

12. Future

2025 AD – end of time



Let's think about health in the future. According to forecasts, average life expectancy in Poland will keep rising. But is a longer life always better? It is also important for life to be healthy and meaningful. Mental health will be just as important as physical health. To stay healthy for years to come, it is worth making time to exercise, sleep well and eat healthily. Personalized medicine is treatment tailored to the patient's genes. Future pandemics will be easier to control thanks to international cooperation and fast vaccines. Heat waves will become more frequent, so cities are creating adaptation plans and cooling centers, and diseases carried by ticks and mosquitoes may become more common because the climate is warming. Noise in cities is expected to be reduced by more electric cars and greenery. Remember these tips – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Noise in the cities of the future is expected to be reduced by...**
 - a) fewer trees
 - b) more car horns
 - c) more highways
 - d) more electric cars and greenery
2. 👁 **According to forecasts, average life expectancy in Poland in the future...**
 - a) will reach 200 years
 - b) will keep rising
 - c) will stay the same
 - d) will drop to 40 years
3. 🗉 **Personalized medicine is treatment that is...**
 - a) done without doctors
 - b) tailored to the patient's genes
 - c) based only on herbs
 - d) the same for everyone
4. 🌡 **Heat waves will become more frequent in the future, so cities are creating...**
 - a) adaptation plans and cooling centers
 - b) less water
 - c) more asphalt
 - d) bans on shade
5. 🏃 **To stay healthy for years to come, it is worth...**
 - a) sit a lot
 - b) exercise, sleep well and eat healthily
 - c) eat plenty of sweets
 - d) skip sleep
6. 🧑 **Future pandemics will be easier to control thanks to...**
 - a) international cooperation and fast vaccines
 - b) ignoring them
 - c) shutting everything down forever
 - d) running away
7. 💖 **In the future, mental health will be just as important as...**
 - a) money
 - b) sports
 - c) fashion
 - d) physical health
8. 🍃 **Diseases carried by ticks and mosquitoes may become more common in Poland because...**
 - a) the climate is warming
 - b) there is more wind
 - c) there is more snow
 - d) there are fewer forests
9. ☆ **Is a longer life always better? It is also important for life to be...**
 - a) only wealthy
 - b) lonely
 - c) healthy and meaningful
 - d) as long as possible at any cost

PHYSICAL CULTURE AND SPORT

12. Future

2025 AD – end of time



Let's look at what the sports of the future may be like. The 2028 Summer Olympics in Los Angeles will include for the first time sports such as cricket and lacrosse, and the 2032 Olympic Games will be held in Brisbane. Poland is bidding to host the Olympic Games in the future. Global warming will make it harder to organize the Winter Olympics, and in hot summers athletes will have to train early in the morning and drink a lot. Esports is competition in video games. Competitions may use AI for judging and motion analysis. Sports will be important because more and more work is done sitting in front of a screen. And will enhancing athletes with technology, for example genetically, be fair? That is a question of the rules of fair play. Remember these plans – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🏁 **The 2032 Olympic Games will be held in...**
 - a) Brisbane
 - b) Moscow
 - c) Warsaw
 - d) Tokyo
2. 🏹 **The 2028 Summer Olympics in Los Angeles will include for the first time sports such as...**
 - a) cricket and lacrosse
 - b) ice hockey
 - c) curling
 - d) ski jumping
3. 🎮 **Esports is competition in...**
 - a) running
 - b) tennis
 - c) swimming
 - d) video games
4. 📅 **In the future, global warming will make it harder to organize...**
 - a) board games
 - b) the Winter Olympics
 - c) the Summer Olympics
 - d) chess tournaments
5. 🤖 **In the future, sports competitions may use AI for...**
 - a) selling tickets
 - b) only counting fans
 - c) replacing athletes
 - d) judging and motion analysis
6. 🌌 **Poland is bidding to host...**
 - a) the ancient Olympics
 - b) a championship on Mars
 - c) competitions in space
 - d) the Olympic Games in the future
7. 💖 **Sports will be important in the future because more and more work is...**
 - a) not done at all
 - b) done while running
 - c) done sitting in front of a screen
 - d) done out in the fields
8. 🍃 **In the hot summers of the future, athletes will have to...**
 - a) avoid drinking water
 - b) wear warm clothes all day
 - c) train at noon
 - d) train early in the morning and drink a lot
9. ⭐ **Will enhancing athletes with technology (e.g., genetically) be fair? This is a question of...**
 - a) the weather
 - b) ticket prices
 - c) fashion
 - d) the rules of fair play

MEDICINE

12. Future

2025 AD – end of time



Let's look into the hospital of the future. Doctors will work together with AI that helps with diagnosis, and AI-powered stethoscopes are meant to detect heart disease by sound. Telesurgery will make it possible to operate from another city using a robot. CRISPR gene therapy may repair faulty genes, and mRNA vaccines may protect against cancers and other diseases. Bacterial resistance to antibiotics is a threat, though – because of it, ordinary infections may become dangerous again. Space medicine must solve the problem of bone loss and radiation. Should genetic treatments be available to everyone? That is a question of fairness in medicine. And anyone can help the medicine of the future, for example by joining a bone marrow donor registry. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **AI-powered stethoscopes of the future are meant to...**
 - a) measure height
 - b) detect heart disease by sound
 - c) play music
 - d) replace doctors
2. 👁 **In the future, mRNA vaccines may protect against...**
 - a) cancers and other diseases
 - b) old age
 - c) broken bones
 - d) only tooth decay
3. 🗨 **In the future, bacterial resistance to antibiotics may mean that...**
 - a) nothing changes
 - b) bacteria disappear
 - c) ordinary infections become dangerous again
 - d) antibiotics become unnecessary
4. ➡ **In the future, CRISPR gene therapy may...**
 - a) replace food
 - b) cure a runny nose
 - c) change eye color for fun
 - d) repair faulty genes
5. 🏠 **Remote surgery (telesurgery) will make it possible to operate...**
 - a) from another city using a robot
 - b) only on animals
 - c) without instruments
 - d) without a doctor
6. 👨 **Doctors of the future will work together with AI that...**
 - a) helps with diagnosis
 - b) decides everything on its own
 - c) fully replaces doctors
 - d) treats without examinations
7. 💖 **Anyone can help the medicine of the future, for example, by...**
 - a) joining a bone marrow donor registry
 - b) ignoring checkups
 - c) avoiding vaccinations
 - d) not going to the doctor
8. 🌿 **Space medicine of the future must solve the problem of...**
 - a) bone loss and radiation
 - b) smog
 - c) traffic jams
 - d) noise
9. ⭐ **Should genetic treatments be available to everyone? This is a question of...**
 - a) fashion
 - b) advertising
 - c) sports
 - d) fairness in medicine

PHARMACY

12. Future

2025 AD – end of time



Let's look into the pharmacy of the future. Medicines designed by AI may be developed much faster than today, and 3D printing of medicines will make it possible to tailor the dose to the patient. Scientists are searching for new antibiotics in soil and oceans, with the help of AI. Phage therapy – treatment with viruses that attack bacteria – may help against resistant bacteria. Biologic drugs made from living cells will become increasingly common. Pharmacies of the future may deliver medicines by drone. Medicinal plants must be protected, because they die out before we study them. Using medicines responsibly means taking them as prescribed. And international programs aim to ensure that essential medicines are available to all, including in poor countries. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🚩 **Medicines of the future designed by AI may be developed...**
 - a) not at all
 - b) only in space
 - c) much faster than today
 - d) more slowly
2. 👁️ **3D printing of medicines will make it possible to...**
 - a) tailor the dose to the patient
 - b) replace doctors
 - c) change the taste of water
 - d) print money
3. 🗨️ **Scientists are searching for the new antibiotics of the future in places such as...**
 - a) clouds
 - b) shops
 - c) soil and oceans, with the help of AI
 - d) cars
4. 🧬 **In the future, biologic drugs (made from living cells)...**
 - a) will disappear
 - b) will be banned
 - c) will be made of plastic
 - d) will become increasingly common
5. 🦠 **Phage therapy – treatment with viruses that attack bacteria – may in the future...**
 - a) replace food
 - b) heal broken bones
 - c) dye hair
 - d) help against resistant bacteria
6. 🚁 **Pharmacies of the future may...**
 - a) deliver medicines by drone
 - b) sell only herbs
 - c) be on the Moon
 - d) be closed
7. 💜 **In the future, using medicines responsibly means...**
 - a) taking them as prescribed
 - b) sharing antibiotics
 - c) taking as many as possible
 - d) throwing them into rivers
8. 🌿 **Medicinal plants must be protected in the future because...**
 - a) they do not work
 - b) they are everywhere
 - c) they die out before we study them
 - d) they are unimportant
9. ⭐ **Should medicines also be available in poor countries? International programs aim to ensure that...**
 - a) essential medicines are available to all
 - b) they are only for the rich
 - c) they are banned
 - d) they disappear

1. c 2. a 3. c 4. d 5. d 6. a 7. a 8. c 9. a

A N S W E R S

AGRICULTURE

12. Future

2025 AD – end of time



Let's head out to the fields of the future. By 2050, to feed about 10 billion people, we will need to produce more and waste less. Drought-resistant crops are being created through breeding and gene editing. Vertical farms are meant to grow vegetables on many floors in cities, and community gardens allow neighbors to grow vegetables together. Agricultural robots may remove weeds with lasers, and farming may use sound, for example to scare away pests. The farmer of the future will use data, drones and sensors. With a warmer climate, farmers in Poland will find it easier to grow grapevines. And lab-grown meat? Supporters say it will reduce animal suffering and emissions. Remember these ideas – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 **The agriculture of the future may use sound, for example...**
 - a) to sing to plants in exchange for fertilizer
 - b) to scare away pests
 - c) to make rain
 - d) to grind grain
2. 👁️ **In the future, vertical farms are meant to...**
 - a) replace all fields
 - b) grow vegetables on many floors in cities
 - c) raise cows in skyscrapers
 - d) grow trees underground
3. 🗨️ **Drought-resistant crops are being created today using methods such as...**
 - a) watering with milk
 - b) breeding and gene editing
 - c) magic spells
 - d) ignoring the problem
4. 📊 **By 2050, to feed about 10 billion people, we will need to...**
 - a) produce more and waste less
 - b) stop eating
 - c) cut down all forests
 - d) eat less than today
5. 🤖 **The agricultural robots of the future may...**
 - a) run shops
 - b) sew clothes
 - c) dance
 - d) remove weeds with lasers
6. 🧑 **Community gardens in cities allow neighbors to...**
 - a) raise elephants
 - b) grow vegetables together
 - c) build parking lots
 - d) argue
7. 💖 **Will you become a farmer of the future? Such a farmer will use...**
 - a) fortune-telling
 - b) data, drones and sensors
 - c) magic
 - d) nothing but a sickle
8. 🌿 **With a warmer climate in the future, farmers in Poland will find it easier to grow...**
 - a) grapevines
 - b) coffee
 - c) coconut palms
 - d) rice in seawater
9. ⭐ **Is lab-grown (cultivated) meat the future? Supporters say that it...**
 - a) tastes better than anything
 - b) grows on trees
 - c) is free
 - d) will reduce animal suffering and emissions

VETERINARY SCIENCE

12. Future

2025 AD – end of time



Let's visit the vet of the future. Collars with sensors may monitor animals' health, and veterinary medicine increasingly uses telemedicine for animals. Care for older dogs and cats will keep improving thanks to medicine, and adopting from shelters may be easier thanks to online databases of animals. Animal diseases may spread farther because the climate is warming and trade is growing. Vaccines for wild animals may protect endangered species from diseases, and cloning endangered animals, such as the black-footed ferret, is meant to help increase the species' genetic diversity. The legal trend in many countries is moving toward better protection of animal welfare. And if you want to become a vet of the future, you need to love animals and study biology. Remember these details – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🎵 In the future, collars with sensors may...
 - a) fully translate dog speech
 - b) feed animals
 - c) play music
 - d) monitor animals' health
2. 👁 In the future, vaccines for wild animals may protect, among others...
 - a) only farmed fish
 - b) plants
 - c) cars
 - d) endangered species from diseases
3. 🗨 Veterinary medicine of the future increasingly uses...
 - a) magic
 - b) fire
 - c) only herbs
 - d) telemedicine for animals
4. 📶 In the future, animal diseases may spread farther because...
 - a) it is getting colder
 - b) the climate is warming and trade is growing
 - c) animals do not travel
 - d) there is no trade anymore
5. 🐾 In the future, care for older dogs and cats...
 - a) will keep improving thanks to medicine
 - b) will be banned
 - c) will not be needed
 - d) will disappear
6. 🐾 In the future, adopting from animal shelters may be easier thanks to...
 - a) a ban on shelters
 - b) online databases of animals
 - c) lotteries
 - d) robots
7. 💖 If you want to become a vet of the future, you need to...
 - a) be afraid of animals
 - b) love animals and study biology
 - c) dislike people
 - d) dislike learning
8. 🌿 Cloning endangered animals, such as the black-footed ferret, is meant to help...
 - a) create new species
 - b) replace wild animals
 - c) increase the species' genetic diversity
 - d) make toys
9. ⭐ Will animals have more rights in the future? The legal trend in many countries is moving toward...
 - a) better protection of their welfare
 - b) banning vets
 - c) treating them as objects
 - d) abolishing protection



ANIMAL SCIENCE AND FISHERIES

12. Future

2025 AD – end of time

Let's look into the farms and fisheries of the future.

Livestock farming is meant to provide animals with welfare and health, and farmers may recognize cows' moods by the sounds they make. Insects as feed and food may replace some meat and fish. Urban beekeeping will develop on rooftops. Aquaculture is meant to be more sustainable and closed-loop. Fisheries must catch fewer fish than can reproduce in time, and marine protected areas will help rebuild fish populations. Warming seas will shift fishing grounds northward, toward colder waters. Should we eat less meat? Scientists point out that it would reduce emissions and water use. Remember these forecasts – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 🦟 In the future, farmers may recognize cows' moods by...
 - a) the sounds they make
 - b) the color of their coat
 - c) their height
 - d) only the smell of their milk
2. 🐟 The aquaculture of the future is meant to be more...
 - a) polluting
 - b) wild
 - c) sustainable and closed-loop
 - d) absent
3. 🦋 In the future, insects as animal feed and human food...
 - a) may replace some meat and fish
 - b) will be banned
 - c) are inedible
 - d) will disappear
4. 🐟 The fisheries of the future must catch fewer fish than...
 - a) there are in aquariums
 - b) can reproduce in time
 - c) the weather allows
 - d) people eat
5. 🐝 In the future, urban beekeeping...
 - a) will disappear
 - b) will develop on rooftops
 - c) will move underground
 - d) will be banned
6. 🐟 In the future, marine protected areas will help...
 - a) ships sail
 - b) rebuild fish populations
 - c) build ports
 - d) catch more fish inside them
7. ❤️ The livestock farming of the future is meant to provide animals with...
 - a) welfare and health
 - b) cramped cages
 - c) darkness
 - d) no movement
8. 🌿 In the future, warming seas will shift fishing grounds...
 - a) into rivers
 - b) northward, toward colder waters
 - c) nowhere
 - d) southward
9. ⭐ Should people eat less meat in the future? Scientists point out that...
 - a) it is impossible
 - b) it would increase emissions
 - c) it would reduce emissions and water use
 - d) it makes no difference

1.a.2.c.3.a.4.b.5.b.6.b.7.a.8.b.9.c

ANSWERS

OTHER (NATURE)

12. Future

2025 AD – end of time



Let's look at nature in the very distant future. Tectonic plates keep moving, so the Himalayas will keep growing. Millions of years from now, East Africa along the East African Rift may split off from the continent. In a few tens of millions of years the Mediterranean Sea may disappear as Africa collides with Europe, and in about 250 million years the continents may merge again into a supercontinent called Pangaea Ultima (Proxima). According to some studies, the next ice age may be delayed by CO₂ emissions. A "library of nature sounds" collects recordings because some species may disappear. Earth Day falls on April 22. The question of what kind of planet we will leave to future generations brings together science, ethics and everyday choices. And for the nature of the future, today you can respect it and teach others to do so. Remember these predictions – I'll ask you about them in a moment.

Read the story aloud. On the back: 9 questions – the answers are in the text.

Place: the field your pawn stands on – link it to the story, or use the app to hear that field's own story.

1. 📌 A “library of nature sounds” collects recordings because in the future...
 - a) some species may disappear
 - b) sounds will be louder
 - c) there will be no microphones
 - d) birds will start talking
2. 👁 In about 250 million years, the continents may merge again into a supercontinent called...
 - a) Atlantis
 - b) New Gondwana
 - c) Pangaea Ultima (Proxima)
 - d) Greater Europe
3. 🗨 In a few tens of millions of years, the Mediterranean Sea may...
 - a) turn into a desert tomorrow
 - b) freeze over
 - c) disappear as Africa collides with Europe
 - d) double in size
4. 📏 Millions of years from now, East Africa along the East African Rift...
 - a) will sink
 - b) will freeze
 - c) will join Asia
 - d) may split off from the continent
5. 🌄 In the future, the Himalayas...
 - a) will start sinking tomorrow
 - b) will keep growing
 - c) will move to Europe
 - d) will disappear
6. 🧊 According to some studies, the next ice age...
 - a) will start in Poland next year
 - b) will certainly never come
 - c) will start tomorrow
 - d) may be delayed by CO₂ emissions
7. ❤ What can you do today for the nature of the future?
 - a) do nothing
 - b) wait
 - c) respect it and teach others to do so
 - d) destroy it
8. 🌿 Both today and in the future, Earth Day falls on...
 - a) November 11
 - b) January 1
 - c) April 22
 - d) December 24
9. ☆ The question “What kind of planet will we leave to future generations?” brings together...
 - a) only scientists
 - b) science, ethics and everyday choices
 - c) no one
 - d) only politicians